Romulus Khalil

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Education

Georgia Institute of Technology - Atlanta, GA — Graduation: Fall 2027
B.S. in Computer Science - Systems/Architecture and Information Internetworks — *Major GPA: 3.6/4.0*

Skills

Languages: C, Rust, Java, Lua, Python, Typescript, PHP
Web/Frameworks: Next.js, Symfony, Nginx, Apache
Tools/DevOps: Kubernetes, Helm, Docker, Linux (Arch, RHEL, Debian, Fedora), AWS (EC2, Bedrock)

Experience

Technology Summer Analyst — June 2026 – Aug. 2026
Morgan Stanley — *Atlanta, GA*

- Migrated and integrated legacy lending services into the firm's credit risk platform, ensuring compatibility across service interfaces and supporting the modernization of backend infrastructure
- Engineered an AI personalization system that analyzes users' chat histories to generate persistent user context, dynamically injecting personalized context into agent system prompts
- Developed agent-level personalization capabilities enabling users to save reusable prompts and generate tailored agent context, integrating configurable agent behavior into the Lending platform
- Contributed to migrating Lending onto a modern agent framework, enabling integration of AI capabilities and accelerating the adoption of agentic workflows across the platform using LangGraph

Undergraduate Teaching Assistant — Aug. 2026 – Present
Design-Operating Systems — *Atlanta, GA*

- Guided 100+ students through core OS concepts: x86 boot procedures, virtual memory management (copy-on-write, zero pages), CPU scheduling, multithreading, filesystems, and networking
- Hold weekly office hours for students and create/grade student exams (2) and student projects (5)

Undergraduate Researcher — Aug. 2025 – Present
Georgia Tech VIP: Configurable Computing/Embedded Systems Team — *Atlanta, GA*

- Developed linux driver for custom 8-pin GPIO module, written in Verilog by the hardware team
- Maximized security by tailoring driver for low-side CPU (prototype has 2 CPUs), protecting the security-critical high-side CPU from malicious button input
- Utilized Xilinx PYNQ-Z1's FPGA for rapid prototyping and testing, creating a comprehensive test suite that covered 100+ use cases

Software Engineering Intern — May 2025 – Aug. 2025
Azalea Health — *Atlanta, GA*

- Created an AI assistant for Azalea Health's Hospital EHR product, leveraging AWS Bedrock for secure access to high performing foundational models like Claude Sonnet and Opus 4.5
- Wrote 12 production-tested PHP APIs used for context/initialization of the Chart Assistant, reducing prompt-to-response time by 50%
- Adopted an agent-based session architecture by spawning one agent per patient chart, dramatically reducing context window overflow and improving responsiveness by ∼3 ms

Projects

The Home Server Project | cribothy.us | *Kubernetes, Debian* — June 2025 – Present

- Manage a multi-node Kubernetes cluster running 20+ containerized services, including media, monitoring, networking, and self-hosted applications
- Implemented GitOps-based infrastructure management with Flux and Kustomize, enabling declarative deployment and automated synchronization from a Git repository
- Configured dynamic persistent storage with an external NFS provisioner, supporting shared RWX volumes for stateful workloads across cluster nodes
- Integrated NVIDIA GPU acceleration into Kubernetes using the NVIDIA Container Toolkit and device plugin, enabling GPU-aware workload scheduling

Coursework

- **Courses:** Design-Operating Systems, Systems and Networking, Computer Organization & Programming, Object Oriented Programming, Data Structures & Algo, Discrete Math, Linear Algebra, Calculus
- **Awards:** Zell Miller Scholarship Recipient, Dean's List 5x